

Intelligent AI-Based Emergency Shelter and Relief Camp Management Platform

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**Scenario-Based Requirement Analysis
CO - 2
Assessment Tool - 2**

Problem Statement

35. Intelligent AI-Based Emergency Shelter and Relief Camp Management Platform

Industry Context

During natural disasters and humanitarian crises, governments and relief organizations establish emergency shelters to accommodate displaced populations. Efficient shelter management is essential for ensuring adequate food, medical care, accommodation, and security. AI, GIS, IoT, cloud

computing, and mobile technologies enable real-time coordination of shelter operations and humanitarian assistance.

Problem Statement

Emergency shelters often experience overcrowding, uneven resource distribution, and poor coordination due to limited real-time information. Manual management methods make it difficult to

monitor occupancy, supplies, and emergency requirements. An AI-based shelter management platform should track shelter occupancy, manage food and medical inventories, allocate resources based on demand, monitor volunteer activities, and generate operational dashboards. The system should improve disaster response while ensuring effective humanitarian assistance.

Objectives

1. Monitor shelter occupancy.
2. Manage relief resources digitally.
3. Optimize resource allocation using AI.
4. Track food and medical supplies.
5. Coordinate volunteers efficiently.
6. Generate shelter management reports.
7. Improve emergency response.
8. Enhance humanitarian service delivery.

Expected Outcomes

1. Efficient shelter operations.
2. Better resource utilization.
3. Faster disaster response.
4. Improved humanitarian assistance.
5. Enhanced volunteer coordination.

6. Increased transparency.

7. Better emergency planning.

8. Improved community resilience.

Intelligent AI-Based Emergency Shelter and Relief Camp Management Platform

Introduction

The Intelligent AI-Based Emergency Shelter and Relief Camp Management Platform is a software solution designed to improve the management of emergency shelters during natural disasters and humanitarian crises. When disasters occur, thousands of people may require temporary shelter, food, medical assistance, and other essential services.

Managing these resources manually often results in delays, overcrowding, inaccurate records, and inefficient distribution of relief materials.

The proposed platform uses Artificial Intelligence (AI), cloud computing, GIS, and modern web technologies to automate shelter operations. It enables authorities to monitor occupancy, manage resources, coordinate volunteers, and generate reports through a centralized system. This architecture design analysis explains the selected software architecture, its components, communication process, quality attributes, and the reasons behind choosing the proposed architectural style.

System Requirement Analysis

System Objectives

The main objective of the proposed system is to provide an intelligent platform that helps disaster management authorities efficiently manage emergency shelters and relief camps. The

system continuously monitors shelter occupancy, manages food and medical inventories, coordinates volunteer activities, and predicts future resource requirements using Artificial Intelligence.

The platform also provides dashboards and reports that support decision-making during emergencies. By automating several manual processes, the system improves response time, reduces operational errors, and ensures better utilization of available resources.

Functional Requirements

The system allows administrators to manage users and shelters, while shelter managers can register residents, update occupancy information, and maintain inventories. Volunteers receive task assignments and update their work status, whereas medical officers monitor medicine availability and treatment records. AI modules analyze data to predict shortages and recommend resource allocation.

Non-Functional Requirements

The platform should provide high availability, strong security, fast response time, and reliable performance. It should support multiple shelters simultaneously while remaining scalable, maintainable, and easy to use.

Quality Attribute Analysis

The software architecture must satisfy several quality attributes to ensure efficient operation during disaster situations.

Scalability

The system should support multiple shelters across different cities or districts without affecting performance. Cloud deployment enables additional computing resources to be added whenever the number of users increases.

Reliability

The platform must operate continuously during emergencies. Automatic backups, fault tolerance, and database replication help prevent data loss and ensure uninterrupted service.

Performance

Real-time occupancy updates and AI predictions require fast processing. Optimized database queries, caching mechanisms, and efficient APIs improve overall system performance.

Security

Sensitive information such as resident records and medical data must be protected using authentication, authorization, encrypted communication, and secure database storage.

Maintainability

The system should be modular so that new features can be added without affecting existing functionality.

Availability

The application should remain accessible at all times because emergency operations continue throughout the day.

Selection of Software Architecture

Selected Architecture

For this project, a **Hybrid Architecture** combining **Layered Architecture** and **Microservices Architecture** is selected.

The Layered Architecture organizes the application into Presentation, Business Logic, AI Services, and Data layers. This separation improves maintainability and simplifies development.

Microservices are used for specialized modules such as AI Prediction, Notification Service, GIS Service, and Reporting Service. These services operate independently and communicate using REST APIs, allowing each component to scale individually.

This combination is highly suitable because disaster management systems require both structured organization and flexible scalability. Independent services reduce system complexity while improving fault isolation and future expansion.

Page 5: Architecture Diagram Description

Overall Architecture

The proposed architecture consists of several layers working together.

Presentation Layer

Users access the system through a web application or mobile application. Different user roles such as Administrator, Shelter Manager, Volunteer, Medical Officer, and Government Officer interact with the platform using secure login credentials.

Application Layer

The application layer processes user requests and implements business rules. It manages shelter registration, occupancy updates, volunteer assignments, inventory management, and report generation.

AI Service Layer

The AI layer predicts shelter occupancy, estimates food and medicine demand, detects shortages, and recommends optimal resource allocation using machine learning models.

Data Layer

The database stores shelter information, resident records, inventory details, volunteer information, AI prediction results, and system logs.

External Services

GIS APIs display shelter locations, while Email and SMS services send emergency notifications to users.

Components and Their Responsibilities

Each architectural component has a specific responsibility within the system.

The **Presentation Layer** provides an easy-to-use interface where users perform daily operations such as registering shelters, updating inventory, viewing dashboards, and checking notifications.

The **Business Logic Layer** validates user requests, applies system rules, and communicates with the database and AI services. It acts as the central processing unit of the application.

The **AI Prediction Service** analyzes historical disaster data and current shelter information to predict future occupancy levels and resource requirements. It helps administrators make informed decisions before shortages occur.

The **Database Server** securely stores all operational data and supports report generation. The **Notification Service** sends alerts whenever occupancy exceeds capacity or inventory falls below a predefined threshold. The **GIS Service** displays shelter locations on digital maps to assist emergency planning.

Component Interaction and Workflow

The workflow begins when an authorized user logs into the system through the web or mobile application. After authentication, the user performs activities such as registering shelters, updating occupancy details, or managing inventories.

These requests are processed by the Business Logic Layer, which validates the information before storing it in the database. Whenever occupancy or inventory data changes, the AI Prediction Service analyzes the updated information and estimates future resource requirements.

If the AI service detects a potential shortage or overcrowding, the Notification Service automatically sends alerts to administrators and shelter managers through email or SMS. Government officials can simultaneously monitor the overall disaster situation using interactive dashboards and reports. This continuous flow of information ensures that decisions are based on real-time data.

Quality Attributes Evaluation

The proposed architecture effectively satisfies important software quality requirements.

The use of microservices allows independent scaling of AI, reporting, and notification modules, ensuring excellent scalability. Security is strengthened through role-based authentication,

encrypted communication, and secure database management. Performance is improved because each service handles a specific responsibility without overloading the entire system.

Reliability is achieved through cloud deployment, backup mechanisms, and database replication, reducing the risk of service interruptions. The layered design makes maintenance easier because modifications in one module have minimal impact on other modules. High availability is ensured through distributed cloud infrastructure, allowing the system to remain operational even if individual services experience failures.

Architectural Decisions and Trade-Off Analysis

The Hybrid Architecture was selected because it combines the advantages of Layered Architecture and Microservices. Layered Architecture provides a clear separation between user interface, business logic, AI processing, and data management. This structure simplifies development and testing.

Microservices provide flexibility by allowing AI prediction, notification, GIS, and reporting modules to operate independently. Individual services can be updated or scaled without affecting the rest of the system.

Although this architecture is more complex than a traditional client-server design, the benefits significantly outweigh the disadvantages. It improves scalability, simplifies future maintenance, and supports integration with external systems. The additional complexity is justified because disaster management systems require high reliability and continuous availability during emergency situations.

Future Enhancements and Conclusion

The proposed architecture is designed to support future expansion without major structural changes. Additional services such as IoT sensor integration, drone-based monitoring, predictive

disease outbreak analysis, weather forecasting APIs, multilingual support, blockchain-based aid tracking, and offline synchronization can be incorporated as independent microservices.

The architecture also supports integration with government disaster management systems and third-party humanitarian organizations through REST APIs. This flexibility makes the platform suitable for large-scale deployments across multiple districts or countries.

Conclusion

The **Intelligent AI-Based Emergency Shelter and Relief Camp Management Platform** requires a reliable, scalable, and secure architecture capable of supporting real-time disaster management operations. The selected Hybrid Architecture effectively separates system responsibilities while allowing independent scaling of critical services. It satisfies important quality attributes such as performance, security, maintainability, reliability, and availability. The architecture provides a strong foundation for future enhancements and enables efficient humanitarian assistance during emergency situations.